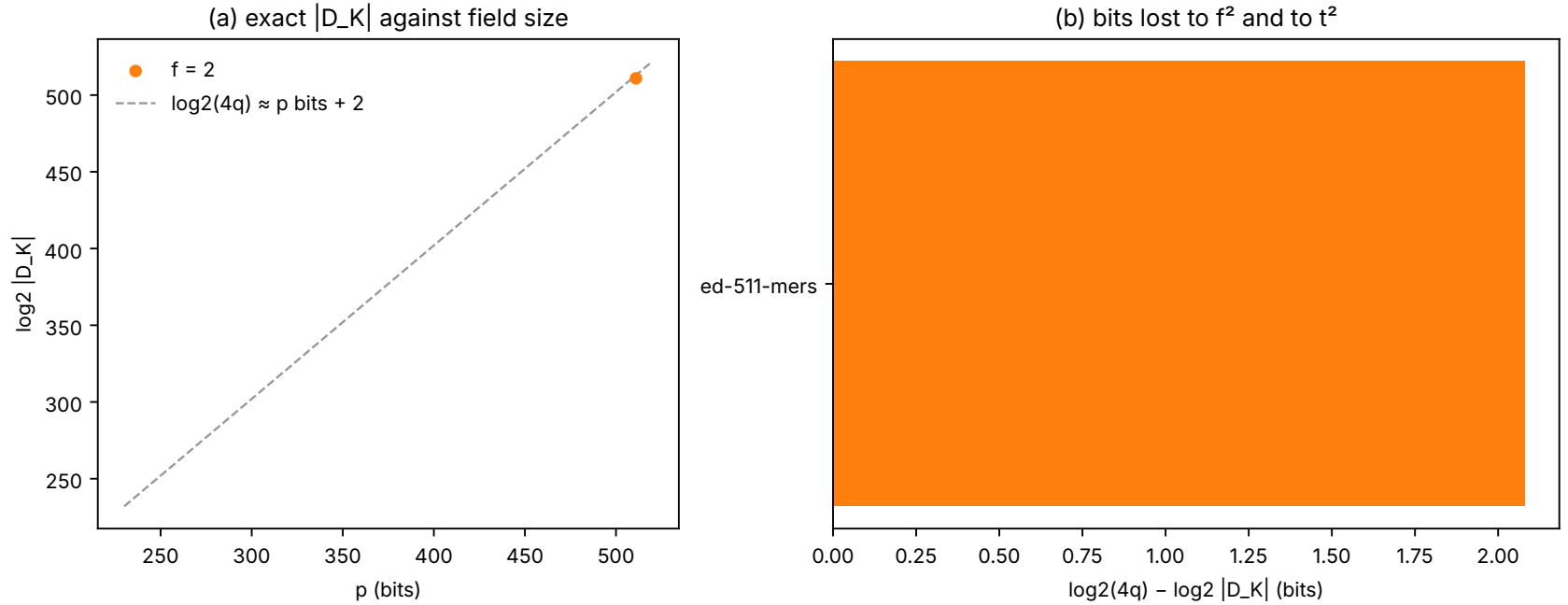
Exact CM discriminant of ed-511-mers (batch 6), and the three composites still open

research/endosweep_dk_exact_batch6_20261105 (2026-11-06). Data: `dk_exact.json`, `factorizations.jsonl`, `open_composites.json`; std-curves commit `77fe6e3585ca2c2225b59d7df24b7c775437276f`. Every factor of $4q - t^2$ is proved prime by PARI `isprime`. These are exact computations, not measurements; the README states method, scope and regeneration.

Part A. nums/ed-511-mers is the one curve completed between the fifth frozen driver snapshot (batch 5) and the sixth (2026-11-06). Its $4q - t^2$ (155 digits) is $2^2 \cdot 11 \cdot p_{21} \cdot p_{47} \cdot p_{86}$. The survey found $2^2 \cdot 11$; GMP-ECM found p_{21} at B1 = 11000 (2026-10-16); this project's driver split the remaining C133 into $p_{47} \cdot p_{86}$ with CADO-NFS 3.0.0, finishing 2026-11-06T13:01:43Z after several resumed runs (the "2658 s wall" in the split log is the last resumed segment only, not the total). So $f = 2$ and $D_K = -s$ with $s \equiv 3 \bmod 4$ (154 digits). std-curves records no `cm_disc`, so D_K was not checked against the literature. The factors were submitted to FactorDB after the split, which now lists the C133 as fully factored.



n = 1 curve; exact integers (PARI isprime-proved factorisations), no uncertainty.

Figure 1: (a) $\log_2|D_K|$ against p for ed-511-mers ($f = 2$). (b) The bits between $\log_2 4q$ and $\log_2|D_K|$. Recorded values from `dk_exact.json`.

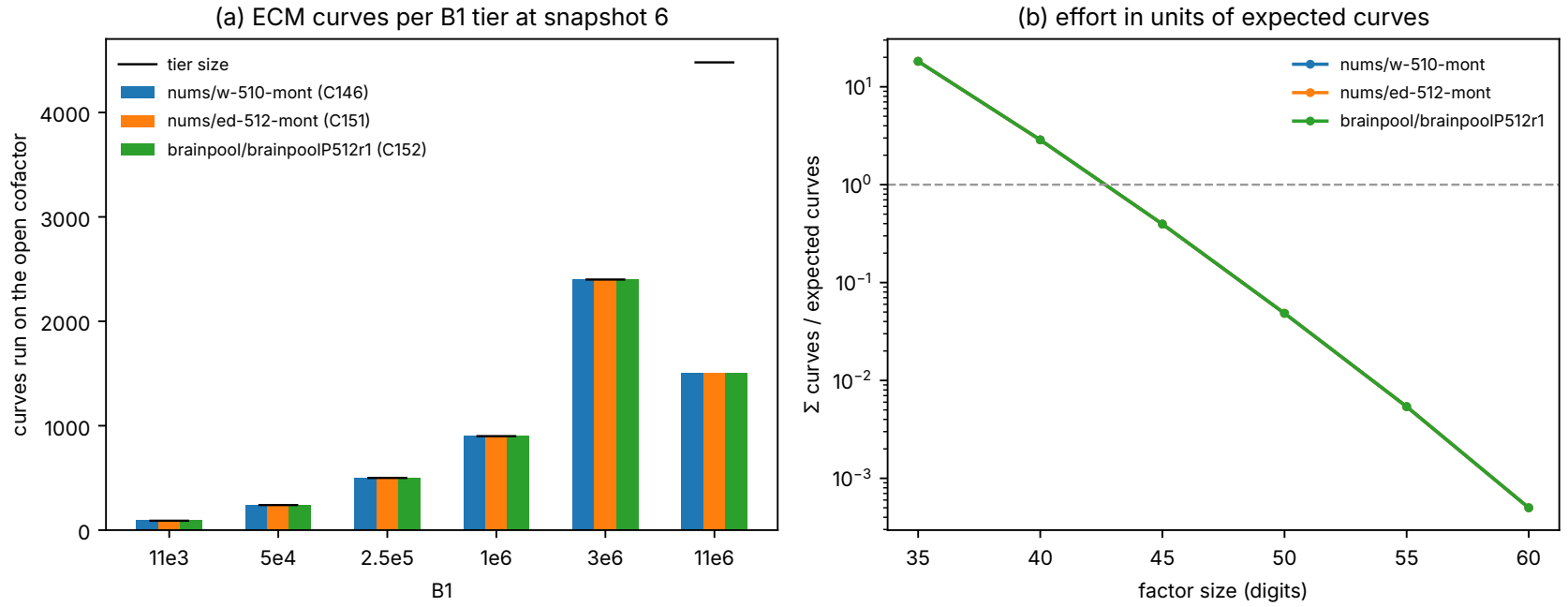
curve	p bits	4q−t² digits	s digits	f	D_K	log2 |D_K|	min degree	std-curves
nums/ed-511-mers	511	155	154	2	(154 digits)	510.92	2^508.92	no cm_disc

Lean: `CryptoResearch.Endosweep.DkExactBatch6.exact_discriminants` (target `TASK-20261106-982b96`), reusing `DkExact.DkCert`; fundamentality is conditional on the primality of the factors above 10^8 (PARI, not Lean).

Part B: the three $4q - t^2$ still open at snapshot 6

No D_K is stated for these. Curves per B1 are those the driver state records; the coverage columns divide them by GMP-ECM 7.0.5′s own expected-curve table (`ecm -v`) and give $\exp(-\sum)$, the approximate chance that a factor of that size, if one exists, was missed (an idealised model). The counts are unchanged from snapshot 5. The C146 to C152 are beyond practical NFS on this 4-core machine.

curve	cofactor	11e3	5e4	2.5e5	1e6	3e6	11e6 (of 4480)	Σ/E 40 dig.	Σ/E 45 dig.	FactorDB
nums/w-510-mont	C146	90	240	500	900	2400	1500	2.87	0.4	C
nums/ed-512-mont	C151	90	240	500	900	2400	1500	2.87	0.4	C
brainpool/brainpoolP512r1	C152	90	240	500	900	2400	1500	2.87	0.4	C



Counts: driver_state.snapshot6.json. Expected curves: GMP-ECM 7.0.5 ecm -v. Dashed: 1 expected find.

Figure 2: (a) ECM curves per B1 tier on each open cofactor at snapshot 6. (b) The same effort in units of GMP-ECM's expected number of curves per factor size; 1 (dashed) is one expected find.